

Lista 03 — DISTINCT, AND-OR-NOT

SQL com SQLite (tabelas PRESERVE e EMPLOYEE)

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Conexão com o banco de dados

```
library(DBI)
library(RSQLite)

con <- dbConnect(RSQLite::SQLite(), "C:/Users/Arthur/Desktop/ME315/Lista_03/me315.sqlite")
```

Hands-On — Lista 3

1. STATE e FEE da tabela PRESERVE, sem duplicados

```
dbGetQuery(con, "
SELECT DISTINCT STATE, FEE
FROM PRESERVE
")
```

	STATE	FEE
1	AZ	3
2	MT	0
3	AZ	0
4	MA	0

2. Valores de DNO na tabela EMPLOYEE, sem duplicados

```
dbGetQuery(con, "  
SELECT DISTINCT DNO  
FROM EMPLOYEE  
")
```

	DNO
1	20
2	10
3	40

3. Comparação entre DNO/SALARY com e sem DISTINCT e com ORDER BY

```
dbGetQuery(con, "SELECT DNO, SALARY FROM EMPLOYEE")
```

	DNO	SALARY
1	20	2000
2	10	2000
3	20	3000
4	40	500
5	10	400
6	20	9000

```
dbGetQuery(con, "SELECT DISTINCT DNO, SALARY FROM EMPLOYEE")
```

	DNO	SALARY
1	20	2000
2	10	2000
3	20	3000
4	40	500
5	10	400
6	20	9000

```
dbGetQuery(con, "SELECT DISTINCT DNO, SALARY FROM EMPLOYEE ORDER BY DNO, SALARY")
```

	DNO	SALARY
1	10	400
2	10	2000
3	20	2000
4	20	3000
5	20	9000
6	40	500

Observação: a primeira consulta mostra as 6 linhas da tabela do jeito que estão, sem tirar repetição. A segunda usa DISTINCT, então se tiver duas linhas com o mesmo par de DNO e SALARY, só uma delas aparece. A terceira é igual a segunda, só que ordenada por DNO e depois por SALARY.

4. Reservas em Montana com menos de 1.000 acres

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND ACRES < 1000
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	3	DANCING PRAIRIE	MT	680	0
2	40	SOUTH FORK MADISON	MT	121	0

5. Reservas com ACRES entre 1200 e 15000 (inclusive)

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE ACRES BETWEEN 1200 AND 15000
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	2	PINE BUTTE SWAMP	MT	15000	0
2	6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

6. Reservas em Montana, sem taxa de entrada, maiores que 10.000 acres

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' AND FEE = 0 AND ACRES > 10000
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	2	PINE BUTTE SWAMP	MT	15000	0

7. Reservas em Montana ou Massachusetts

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR STATE = 'MA'
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	3	DANCING PRAIRIE	MT	680	0
2	40	SOUTH FORK MADISON	MT	121	0
3	14	MCELWAIN-OLSEN	MA	66	0
4	13	TATKON	MA	40	0
5	9	DAVID H. SMITH	MA	830	0
6	11	MIACOMET MOORS	MA	4	0
7	12	MOUNT PLANTAIN	MA	730	0
8	1	COMERTOWN PRAIRIE	MT	1130	0
9	2	PINE BUTTE SWAMP	MT	15000	0
10	10	HOFT FARM	MA	90	0

8. Reservas em Montana ou com menos de 1.000 acres

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE STATE = 'MT' OR ACRES < 1000
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	5	HASSAYAMPA RIVER	AZ	660	3
2	3	DANCING PRAIRIE	MT	680	0
3	40	SOUTH FORK MADISON	MT	121	0
4	14	MCELWAIN-OLSEN	MA	66	0
5	13	TATKON	MA	40	0
6	9	DAVID H. SMITH	MA	830	0
7	11	MIACOMET MOORS	MA	4	0
8	12	MOUNT PLANTAIN	MA	730	0
9	1	COMERTOWN PRAIRIE	MT	1130	0
10	2	PINE BUTTE SWAMP	MT	15000	0
11	80	RAMSEY CANYON	AZ	380	3
12	10	HOFT FARM	MA	90	0

9. Número e nome de reservas com taxa de entrada diferente de zero

```
dbGetQuery(con, "
SELECT PNO, PNAME
FROM PRESERVE
WHERE FEE <> 0
")
```

	PNO	PNAME
1	5	HASSAYAMPA RIVER
2	80	RAMSEY CANYON
3	6	PAPAGONIA-SONOITA CREEK

10. Número e nome de reservas sem taxa de US\$ 3,00 e sem taxa de US\$ 10,00

```
dbGetQuery(con, "
SELECT PNO, PNAME
FROM PRESERVE
WHERE FEE <> 3 AND FEE <> 10
")
```

	PNO	PNAME
1	3	DANCING PRAIRIE
2	7	MULESHOE RANCH

3	40	SOUTH FORK MADISON
4	14	MCELWAIN-OLSEN
5	13	TATKON
6	9	DAVID H. SMITH
7	11	MIACOMET MOORS
8	12	MOUNT PLANTAIN
9	1	COMERTOWN PRAIRIE
10	2	PINE BUTTE SWAMP
11	10	HOFT FARM

11. Reservas no Arizona sem taxa, ou qualquer reserva com menos de 100 acres

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE (STATE = 'AZ' AND FEE = 0) OR ACRES < 100
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	7	MULESHOE RANCH	AZ	49120	0
2	14	MCELWAIN-OLSEN	MA	66	0
3	13	TATKON	MA	40	0
4	11	MIACOMET MOORS	MA	4	0
5	10	HOFT FARM	MA	90	0

12. Reservas com menos de 1.000 acres e taxa zero, ou localizadas no Arizona

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE (ACRES < 1000 AND FEE = 0) OR STATE = 'AZ'
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	5	HASSAYAMPA RIVER	AZ	660	3
2	3	DANCING PRAIRIE	MT	680	0
3	7	MULESHOE RANCH	AZ	49120	0
4	40	SOUTH FORK MADISON	MT	121	0

5	14	MCELWAIN-OLSEN	MA	66	0
6	13	TATKON	MA	40	0
7	9	DAVID H. SMITH	MA	830	0
8	11	MIACOMET MOORS	MA	4	0
9	12	MOUNT PLANTAIN	MA	730	0
10	80	RAMSEY CANYON	AZ	380	3
11	10	HOFT FARM	MA	90	0
12	6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

13. Reservas sem taxa de entrada (FEE não maior que zero), ou em Montana com mais de 1.000 acres

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE NOT FEE > 0 OR (STATE = 'MT' AND ACRES > 1000)
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	3	DANCING PRAIRIE	MT	680	0
2	7	MULESHOE RANCH	AZ	49120	0
3	40	SOUTH FORK MADISON	MT	121	0
4	14	MCELWAIN-OLSEN	MA	66	0
5	13	TATKON	MA	40	0
6	9	DAVID H. SMITH	MA	830	0
7	11	MIACOMET MOORS	MA	4	0
8	12	MOUNT PLANTAIN	MA	730	0
9	1	COMERTOWN PRAIRIE	MT	1130	0
10	2	PINE BUTTE SWAMP	MT	15000	0
11	10	HOFT FARM	MA	90	0

14. Todas as reservas, exceto as de Montana sem taxa de entrada

```
dbGetQuery(con, "
SELECT *
FROM PRESERVE
WHERE NOT (STATE = 'MT' AND FEE = 0)
")
```

	PNO	PNAME	STATE	ACRES	FEE
1	5	HASSAYAMPA RIVER	AZ	660	3
2	7	MULESHOE RANCH	AZ	49120	0
3	14	MCELWAIN-OLSEN	MA	66	0
4	13	TATKON	MA	40	0
5	9	DAVID H. SMITH	MA	830	0
6	11	MIACOMET MOORS	MA	4	0
7	12	MOUNT PLANTAIN	MA	730	0
8	80	RAMSEY CANYON	AZ	380	3
9	10	HOFT FARM	MA	90	0
10	6	PAPAGONIA-SONOITA CREEK	AZ	1200	3

Equivalências lógicas (AND / OR / NOT)

15.

```
WHERE STATE = 'MA' AND (ACRES > 1000 OR FEE = 0.0)
WHERE (STATE = 'MA' AND ACRES > 1000) OR (STATE = 'MA' AND FEE = 0.0)
```

São equivalentes. É a propriedade distributiva: $C1 \text{ AND } (C2 \text{ OR } C3) = (C1 \text{ AND } C2) \text{ OR } (C1 \text{ AND } C3)$.

16.

```
WHERE STATE = 'MA' OR (ACRES > 1000 AND FEE = 0.0)
WHERE (STATE = 'MA' OR ACRES > 1000) AND (STATE = 'MA' OR FEE = 0.0)
```

São equivalentes. É a propriedade distributiva: $C1 \text{ OR } (C2 \text{ AND } C3) = (C1 \text{ OR } C2) \text{ AND } (C1 \text{ OR } C3)$.

17.

```
WHERE NOT (ACRES < 50 AND STATE = 'MA')
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
WHERE ACRES >= 50 OR STATE <> 'MA'
```

Pela Lei de De Morgan, $\text{NOT } (C1 \text{ AND } C2) = (\text{NOT } C1) \text{ OR } (\text{NOT } C2)$. Então a 1ª e a 3ª cláusula são equivalentes entre si, porque as duas usam OR. Já a 2ª cláusula não é equivalente às outras, porque ela usa AND no lugar de OR.

18.

```
WHERE NOT (ACRES < 50 OR STATE = 'MA')  
WHERE NOT ACRES < 50 OR NOT STATE = 'MA'  
WHERE NOT ACRES < 50 AND NOT STATE = 'MA'
```

Pela Lei de De Morgan, $\text{NOT } (C1 \text{ OR } C2) = (\text{NOT } C1) \text{ AND } (\text{NOT } C2)$. Então a 1ª e a 3ª cláusula são equivalentes entre si, porque as duas usam AND. Já a 2ª cláusula não é equivalente às outras, porque ela usa OR no lugar de AND.

```
dbDisconnect(con)
```

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

```
[1] "2026-09-01 00:08:53 -03"
```